4. $X_1 \cdots X_m \overset{iid}{\sim} N(\mu_1, \sigma^2)$, $Y_1 \cdots Y_n \overset{iid}{\sim} N(\mu_2, \sigma^2)$, X_i, Y_i indep.

未知：μ_1, μ_2, σ^2

sample mean : $\bar{X} = \frac{1}{m}\sum_{i=1}^{m} X_i$, $\bar{Y} = \sum_{i=1}^{n} Y_i$

sample variance : $S_X^2 = \frac{1}{m-1}\sum_{i=1}^{m}(X_i - \bar{X})^2$, $S_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$

(i) joint density $f = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^m \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{m}(x_i-\mu_1)^2\right\} \cdot \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left\{-\frac{1}{2\sigma^2}\sum_{j=1}^{n}(y_j-\mu_2)^2\right\}$

$= (2\pi\sigma^2)^{-\frac{m+n}{2}} \exp\left\{-\frac{1}{2\sigma^2}\left(\sum_{i=1}^{m}(x_i-\mu_1)^2 + \sum_{j=1}^{n}(y_j-\mu_2)^2\right)\right\}$

log likelihood function $L = -\frac{m+n}{2}\log 2\pi\sigma^2 - \frac{1}{2\sigma^2}\left(\sum_{i=1}^{m}(x_i-\mu_1)^2 + \sum_{j=1}^{n}(y_j-\mu_2)^2\right)$

$\frac{\partial L}{\partial \mu_1} = -\frac{1}{2\sigma^2}\cdot(-2)\sum_{i=1}^{m}(x_i-\mu_1) = \frac{1}{\sigma^2}\sum_{i=1}^{m}(x_i-\mu_1) = 0 \Rightarrow \hat{\mu}_1 = \bar{X}$

同理，$\hat{\mu}_2 = \bar{Y}$

$\frac{\partial L}{\partial \sigma^2} = -\frac{m+n}{2}\frac{1}{\sigma^2} + \frac{1}{2\sigma^4}\left(\sum_{i=1}^{m}(x_i-\mu_1)^2 + \sum_{j=1}^{n}(y_i-\mu_2)^2\right) = 0 \Rightarrow \hat{\sigma}^2 = \frac{\sum_{i=1}^{m}(x_i-\mu_1)^2 + \sum_{j=1}^{n}(y_i-\mu_2)^2}{m+n} = \frac{mS_X^2 + nS_Y^2}{m+n}$

接下来看 UMVUE，找充分完备统计量

$f = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^m \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{m}(x_i-\mu_1)^2\right\} \cdot \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left\{-\frac{1}{2\sigma^2}\sum_{j=1}^{n}(y_j-\mu_2)^2\right\}$

$(\bar{X}, \bar{Y}, (m-1)S_X^2 + (n-1)S_Y^2)$ 是 (μ_1, μ_2, σ^2) 的 complete and sufficient statistics

$E(\bar{X}) = \mu_1$, $E(\bar{Y}) = \mu_2$

$E((m-1)S_X^2 + (n-1)S_Y^2) = E((m-1)S_X^2) + E((n-1)S_Y^2)$

$\frac{(m-1)S_X^2}{\sigma^2} \sim \chi^2_{m-1}$, $\frac{(n-1)S_Y^2}{\sigma^2} \sim \chi^2_{n-1}$

$= (m-1)\sigma^2 + (n-1)\sigma^2 = (m+n-2)\sigma^2$

无偏估计量为 $\frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}$, by L-S theorem , it's UMVUE for σ^2

(ii) $T = \frac{\sqrt{n}(\bar{X}-\mu_1) + \sqrt{m}(\bar{Y}-\mu_2)}{\sqrt{(m-1)S_X^2 + (n-1)S_Y^2}}$

$\bar{X} \sim N(\mu_1, \frac{\sigma^2}{m})$, $\bar{Y} \sim N(\mu_2, \frac{\sigma^2}{n})$

$\bar{X} - \mu_1 \sim N(0, \frac{\sigma^2}{m})$, $\bar{Y} - \mu_2 \sim N(0, \frac{\sigma^2}{n})$

$\alpha(\bar{X}-\mu_1) \sim N(0, \frac{\alpha^2\sigma^2}{m})$, $\beta(\bar{Y}-\mu_2) \sim N(0, \frac{\beta^2\sigma^2}{n})$

$\alpha(\bar{X}-\mu_1)+\beta(\bar{Y}-\mu_2) \sim N(0, (\frac{\alpha^2}{m}+\frac{\beta^2}{n})\sigma^2) \Rightarrow \frac{\alpha(\bar{X}-\mu_1)+\beta(\bar{Y}-\mu_2)}{\sqrt{(\frac{\alpha^2}{m}+\frac{\beta^2}{n})\sigma^2}} \sim N(0,1)$ ($\bar{X} \perp\!\!\!\perp \bar{Y}$)

$\frac{(m-1)S_X^2}{\sigma^2} \sim \chi^2_{m-1}$, $\frac{(n-1)S_Y^2}{\sigma^2} \sim \chi^2_{n-1} \Rightarrow \frac{(m-1)S_X^2+(n-1)S_Y^2}{\sigma^2} \sim \chi^2_{m+n-2}$ ($\bar{X} \perp\!\!\!\perp \bar{Y}$)

t分布 : $\frac{Z}{\sqrt{\frac{\chi}{df}}} \sim t_{df}$, with $Z \perp\!\!\!\perp \chi^2$

$$\frac{\frac{\alpha(\bar{X}-\mu_1)+\beta(\bar{Y}-\mu_2)}{\sqrt{(\frac{\alpha^2}{m}+\frac{\beta^2}{n})\sigma^2}}}{\sqrt{\frac{(m-1)S_X^2+(n-1)S_Y^2}{\sigma^2(m+n-2)}}} = \frac{\alpha(\bar{X}-\mu_1)+\beta(\bar{Y}-\mu_2)}{\sqrt{(m-1)S_X^2+(n-1)S_Y^2}} \cdot \frac{\sqrt{m+n-2}}{\sqrt{\frac{\alpha^2}{m}+\frac{\beta^2}{n}}} \sim t_{n+m-2}$$

i.e. $T \sim \sqrt{\frac{\frac{\alpha^2}{m}+\frac{\beta^2}{n}}{m+n-2}}\, t_{n+m-2}$

5. $y_i = \beta x_i + \varepsilon_i$

x_i : fixed ; $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

(i) $y_i \sim N(\beta x_i, \sigma^2)$

$f(y_1 \cdots y_n) = (2\pi\sigma^2)^{-\frac{n}{2}} \cdot \exp\{-\frac{1}{2\sigma^2}\sum_{i=1}^n (y_i-\beta x_i)^2\}$

log likelihood function : $L = -\frac{n}{2}\log 2\pi\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n (y_i-\beta x_i)^2$

$\frac{\partial L}{\partial \beta} = -\frac{1}{2\sigma^2}\sum_{i=1}^n 2(y_i-\beta x_i)(-x_i) = \frac{1}{\sigma^2}\sum_{i=1}^n 2x_i(y_i-\beta x_i) = 0 \Rightarrow \hat{\beta}_{MLE} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$

$\frac{\partial L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^n (y_i-\beta x_i)^2 = 0 \Rightarrow \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^n (y_i-\hat{\beta}x_i)^2$

(ii) $\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$ 其中 $y_i \sim N(\beta x_i, \sigma^2)$ 且 $y_1 \cdots y_n$ indep.

$x_i y_i \sim N(\beta x_i^2, x_i^2\sigma^2)$ indep.

$\sum_{i=1}^n x_i y_i \sim N(\beta \sum_{i=1}^n x_i^2, \sum_{i=1}^n x_i^2\sigma^2)$

$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} \sim N(\beta, \frac{\sigma^2}{\sum_{i=1}^n x_i^2})$

$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n (y_i-\hat{\beta}x_i)^2$ $\hat{\beta}x_i \sim N(x_i\beta, \frac{x_i\sigma^2}{\sum_{i=1}^n x_i^2})$

$y_i \sim N(x_i\beta, \sigma^2)$

$y_i - \hat{\beta}x_i \not\sim N(0, (\frac{x_i}{\sum_{i=1}^{n} x_i^2} + 1)\sigma^2)$, since y_i and $\hat{\beta}x_i$ are not independent

一元线性回归的矩阵表述

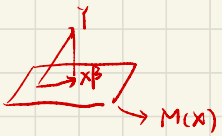
$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \beta + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix} \quad , \quad Y_{n\times 1} = X_{n\times 1}\beta + \varepsilon_{n\times 1}$$

maximize likelihood function $\Leftrightarrow$ minimize $\sum_{i=1}^{n}(y_i - x_i\beta)^2$

$$\frac{\partial |Y - x\beta|^2}{\partial \beta} = x^T(x\beta - Y) = 0 \Rightarrow x^T x\beta = x^T Y$$

$$\hat{\beta} = (x^T x)^{-1} x^T Y$$

Note : $x\hat{\beta}$ is the best predictor of Y on $M(x)$



$Y \sim N(x\beta, \sigma^2 I)$ so $\hat{\beta} \sim N(\beta, (x^T x)^{-1} x^T \cdot \sigma^2 I \cdot ((x^T x)^{-1} x^T)^T)$

$$\hat{\beta} \sim N(\beta, \sigma^2 (x^T x)^{-1})$$

$$\hat{\varepsilon} = Y - x\hat{\beta} = Y - x(x^T x)^{-1} x^T Y = Y(I - x(x^T x)^{-1} x^T)$$

projection matrix , denote as H

then $\hat{\varepsilon} = (I - H)Y$

$$\hat{\sigma}^2 = \frac{1}{n-1}\hat{\varepsilon}^T\hat{\varepsilon} = \frac{1}{n-1} Y^T(I-H)^T(I-H)Y = \frac{1}{n-1}\varepsilon^T(I-H)\varepsilon$$

$(I-H)$ is idempotent matrix

Also , the df of $\hat{\sigma}^2$ is the trace of $I-H$

$$tr(I-H) = tr(I) - tr(H)$$

$$= tr(I) - tr(x(x^T x)^{-1} x^T)$$

$$= n - tr((x^T x)^{-1} x^T x) = n-1 \Rightarrow \hat{\sigma}^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}$$

(iii) $\frac{1}{n-1} Y^T(I-H)Y \perp\!\!\!\perp (x^T x)^{-1} x^T Y$, only needs to show $(I-H)(x^T x)^{-1} x^T = 0$

$$(I - x(x^T x)^{-1} x^T)((x^T x)^{-1} x^T) = (x^T x)^{-1} x^T - (x^T x)^{-1} x^T = 0$$